

MOBILE EXPENSE TRACKER WITH BUDGET ANOMALY DETECTION

Authors: VILLAFUERTE, KYLA A. | RAMOS, JOHN G.

Year: 2024

Department of Information Technology

Degree Program: BSIT

Keywords: mobile app, expense tracking, anomaly detection, Isolation Forest, Flutter

ABSTRACT

This study develops a Flutter-based mobile expense tracking application with Isolation Forest anomaly detection to flag unusual spending patterns, achieving 89.4% anomaly detection accuracy in user trials.

INTRODUCTION

Personal financial management is a challenge for Filipino college students, many of whom lack formal budgeting skills. Mobile expense trackers exist but typically lack intelligent analysis. This study adds anomaly detection to a mobile expense tracker to proactively alert users when spending deviates significantly from their historical patterns, supporting financial literacy.

METHODOLOGY

A Flutter mobile application was developed for Android and iOS. The backend uses FastAPI (Python) with SQLite. Expense data is categorized into 8 categories (Food, Transport, Academic, etc.). Isolation Forest is applied on rolling 30-day spending vectors to detect outliers (contamination=0.1). The app was beta-tested with 40 CvSU students over 60 days. Evaluation: precision and recall of anomaly alerts vs. self-reported unexpected expenses.

RESULTS

Isolation Forest achieved 89.4% precision and 84.7% recall for anomaly detection against user self-reports. Users received an average of 2.3 alerts per month. 78% of users rated alerts as 'useful' or 'very useful'. Average session time increased 22% vs. a control app without anomaly alerts. App crash rate was 0.3% over 60 days.

DISCUSSION

Anomaly detection meaningfully enhances expense tracking utility by providing proactive insights rather than passive logging. False positives (10.6%) were mainly triggered by irregular but legitimate expenses like semester fees. Future work should allow user-defined category budgets as anomaly baselines and integrate with GCash/Maya transaction history APIs.